George Wanis's report for milestone one after studying the playlist by Michel van Biezen

A 4-state Kalman filter (px, py, vx, vy) was run for one predict + correct step in four scenarios. Each plot shows the prediction (blue, dashed), the measurement (red, dotted) and the correction (green, solid) as 1D Gaussians N(mean, variance) for one state variable.

- Prediction mean = previous state + control.
With dt = 1 s: px = 0 + 10 + 0.5*2 = 11 m, vx = 10 + 2 = 12 m/s
(scenarios 1-3); nothing moves in scenario 4 because x0 and u are zero.

- Prediction variance = P0 + Q, and for position it also includes the velocity variance (a velocity error becomes a
position error over dt).
So position variance = 1.0 + 1.0 + Qpos, velocity variance = 1.0 + Qvel.

- Measurement is centred on the sensor reading; its width is exactly the sensor variance R (GPS for position,speedometer for velocity).

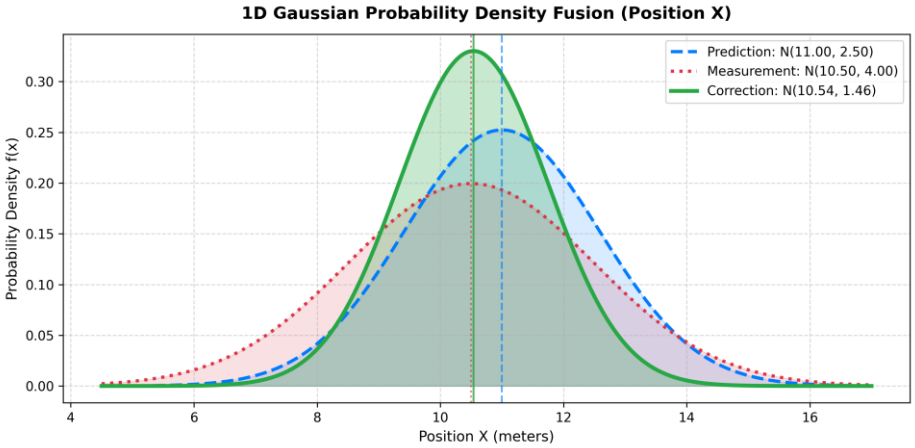
- Correction is a variance-weighted blend: the mean moves toward whichever of the two is narrower, and the corrected variance is always smaller than both inputs, because two sources of information are better than one.

- Because the prediction links position and velocity errors (non-zero covariance), the full 4D update lets the velocity innovation nudge position and vice versa. Corrected means therefore differ slightly from a textbook independent 1D fusion; this is called out where it matters.

Scenario 1 - Noisy Prediction

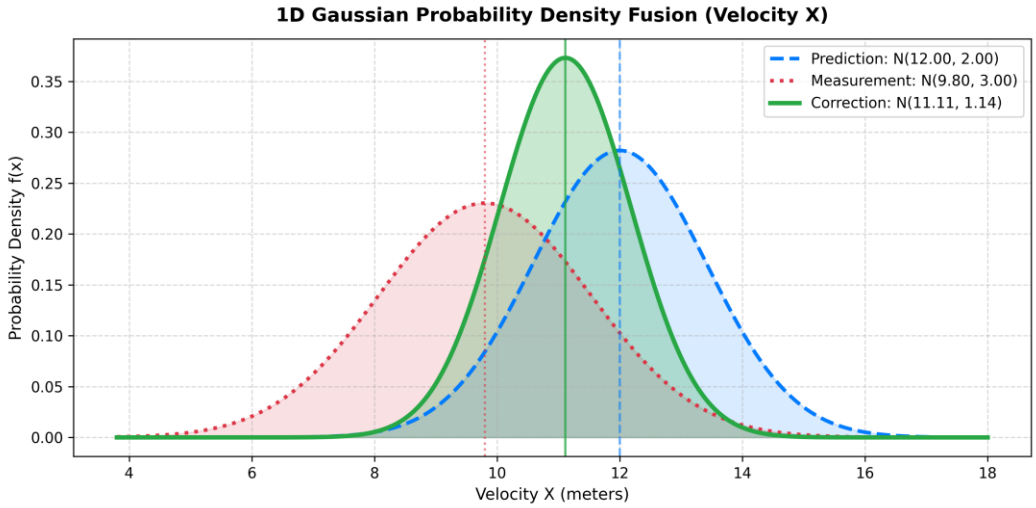
Q: pos 0.5, vel 1.0 (large) | R: GPS 4.0, speedometer 3.0 | z = [10.5, 5.2, 9.8, 4.9]

Position X



- **Prediction:** N(11, 2.5). Mean from motion model (0 + 10 + 1). The large process noise (0.5) plus the velocity uncertainty (1.0) inflates the variance well above the initial 1.0.
- **Measurement:** N(10.5, 4.0). The GPS is 0.5 m below the prediction and wider than it (R = 4).
- **Correction:** N(10.54, 1.46). The result sits close to the (narrower) prediction but is pulled slightly toward the GPS; variance drops from 2.5 to 1.46.
- **Parameters:** the large Q makes the prediction only moderately more trustworthy than the sensor, so the filter gives the measurement a substantial weight (about 40%).

Velocity X

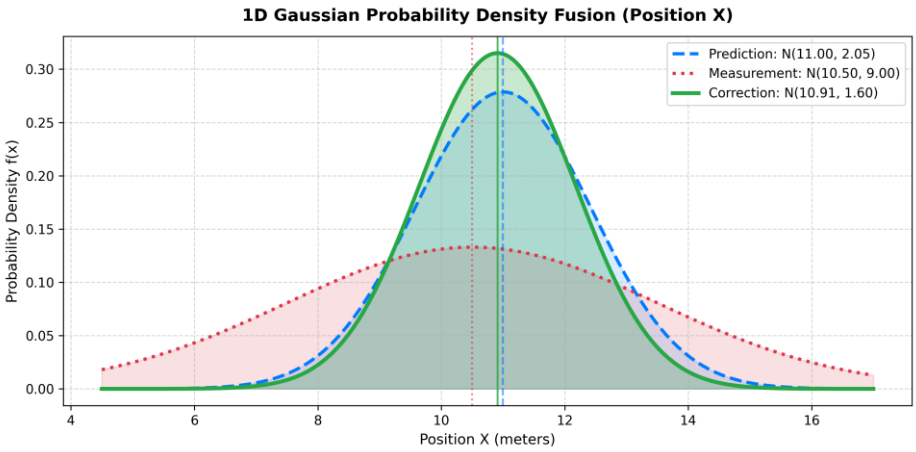


- **Prediction:** N(12, 2.0): mean 10 + 2*1; variance 1.0 (P0) + 1.0 (Q).
- **Measurement:** N(9.8, 3.0). The speedometer disagrees with the prediction by 2.2 m/s.
- **Correction:** N(11.11, 1.14). The estimate moves about 0.9 m/s toward the measurement, closer to the narrower prediction.
- **Parameters:** the big velocity process noise (1.0) widens the prediction, so the filter listens to the speedometer more than it would with a small Q.

Scenario 2 - Noisy Measurement

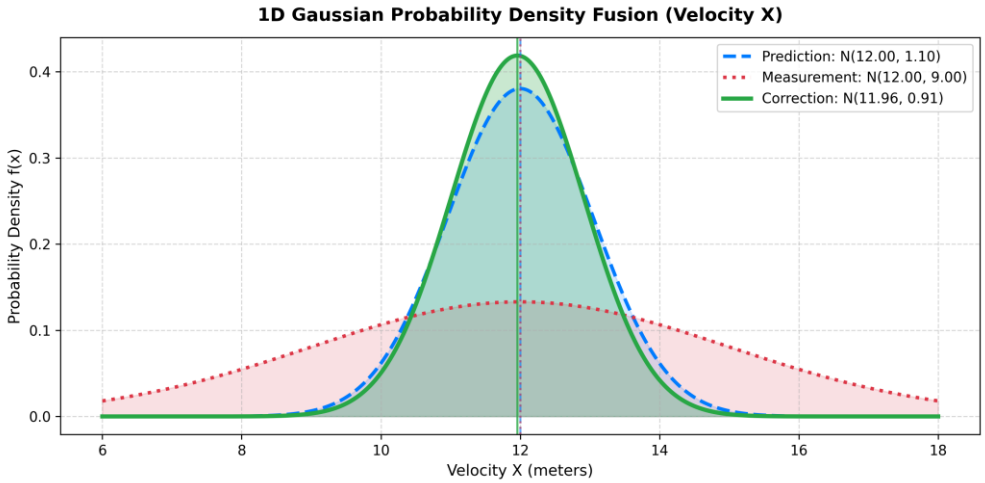
Q: pos 0.05, vel 0.1 (small) | R: GPS 9.0, speedometer 9.0 (large) | z = [10.5, 5.2, 12.0, 6.0]

Position X



- **Prediction:** N(11, 2.05). Same mean as scenario 1, but the small Q gives a tighter distribution (2.05 vs 2.5).
- **Measurement:** N(10.5, 9.0). Very wide and flat: the GPS is declared unreliable (std 3 m).
- **Correction:** N(10.91, 1.60). The estimate stays close to the prediction (moved only 0.09 m); the filter trusts its
model about 4x more than the GPS.
- **Parameters:** lowering Q and raising R both shift trust toward the prediction. Compared with scenario 1, the corrected mean is closer to 11 and the measurement contributes less.

Velocity X

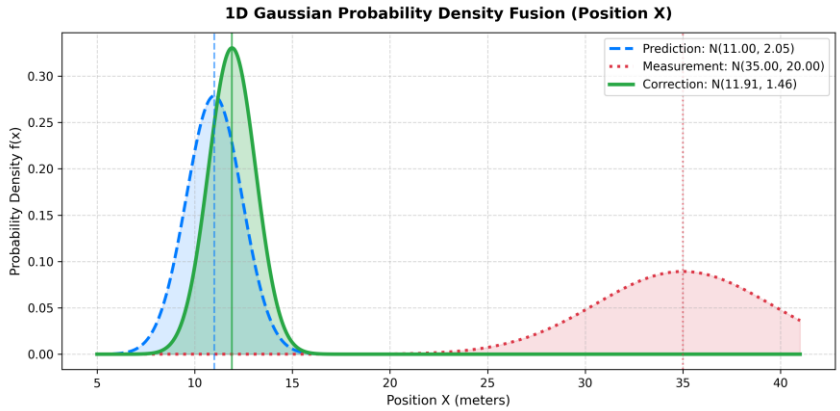


- **Prediction:** N(12, 1.1) - narrow, since Qvel is only 0.1.
- **Measurement:** N(12.0, 9.0) - centred exactly on the prediction but very wide.
- **Correction:** N(11.96, 0.91). The means coincide so the estimate barely moves (the -0.04 shift comes from the position innovation leaking in through the position-velocity correlation).
Variance still shrinks from 1.1 to 0.91: even a noisy measurement that agrees adds information.
- **Parameters:** agreement between sources changes the variance, not the mean; the amount of shrinkage is set by R.

Scenario 3 - GPS Outlier Glitch

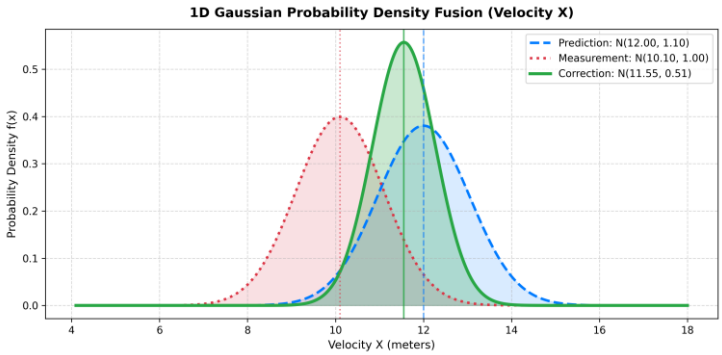
Q: pos 0.05, vel 0.1 | R: GPS 20.0 (very noisy), speedometer 1.0 (precise) | z = [35, 28, 10.1, 5.0]

Position X



- **Prediction:** N(11, 2.05), same as scenario 2.
- **Measurement:** N(35, 20). The GPS reads 24 m away from the prediction, roughly 5 standard deviations of the combined uncertainty - clearly a glitch. Its wide curve (R = 20) is the filter's only protection.
- **Correction:** N(11.91, 1.46). The estimate is essentially unchanged (moved ~0.9 m instead of 24 m). Fusing the GPS alone would move it ~2.2 m; the speedometer's disagreement partly cancels that through the correlation.
- **Parameters:** declaring a large R for the GPS is what keeps the outlier from dragging the track. Note a standard Kalman filter has no outlier rejection: the glitch still has some effect, it is only down-weighted.

Velocity X

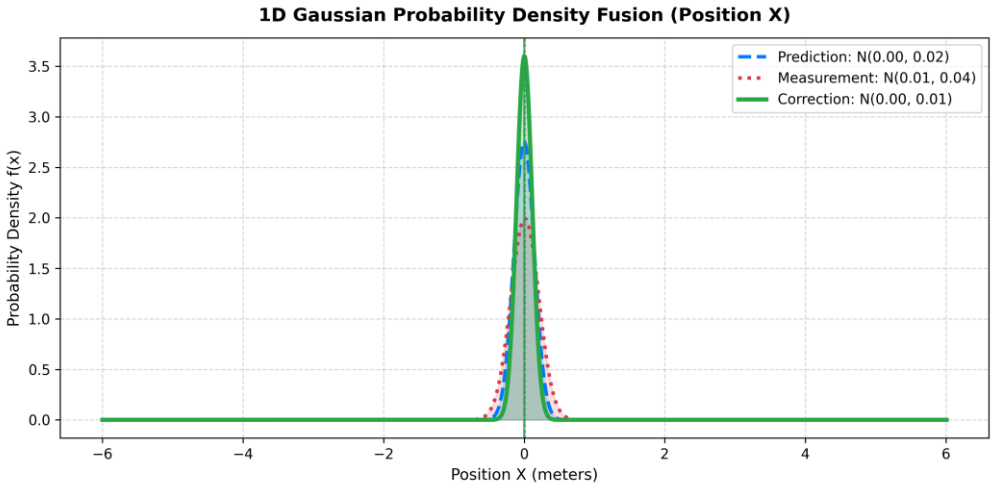


- **Prediction:** N(12, 1.1).
- **Measurement:** N(10.1, 1.0). The precise speedometer (R = 1) is a narrow curve of comparable width to the prediction.
- **Correction:** N(11.55, 0.51). The estimate lands between the two and variance is roughly halved, because both sources are confident. Independent fusion would give ~11.0; the extra +0.55 is the GPS glitch leaking into velocity through the correlation.
- **Parameters:** a good sensor (small R) dominates; a bad sensor on one quantity can still slightly contaminate another when the state is correlated.

Scenario 4 - Cold Start (Docked)

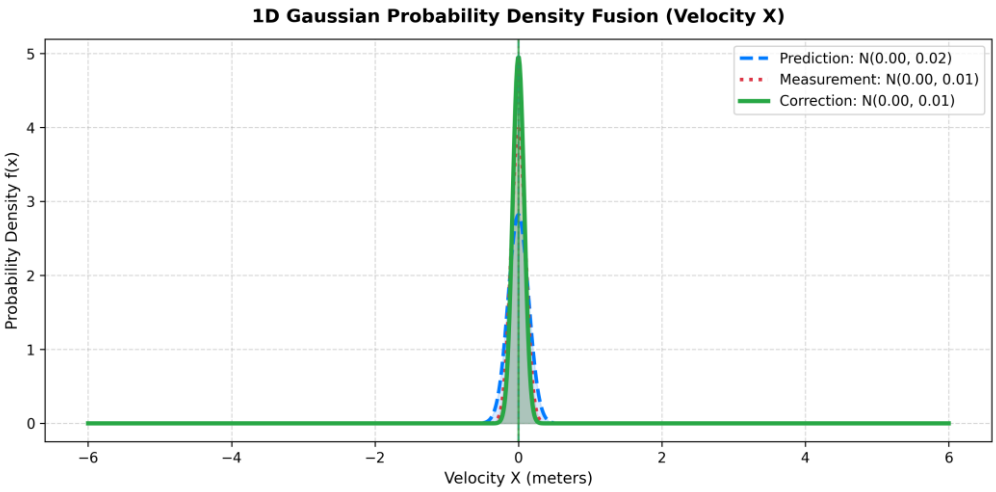
x0 = 0, P0 = 0.01, u = 0 | Q: pos 0.001, vel 0.01 | R: GPS 0.04, speedometer 0.01 | z = [0.01, 0, 0, 0]

Position X



- **Prediction:** N(0, 0.021). The vehicle is stationary with no acceleration, so the mean stays at 0; the tiny variance comes from P0 (0.01) + velocity variance (0.01) + Q (0.001).
- **Measurement:** N(0.01, 0.04). The GPS reads 1 cm off, well within its 0.2 m standard deviation.
- **Correction:** N(0.003, 0.012). The mean moves only ~3 mm; variance falls to about 0.012.
- **Parameters:** everything is small, so the plot is a narrow spike. The visualizer uses a fixed +/-6 m window, which is why the curves look thin; the relative picture (green narrowest) is what matters.

Velocity X



- **Prediction:** N(0, 0.02) - from P0 (0.01) + Qvel (0.01).
- **Measurement:** N(0, 0.01). The speedometer reads exactly 0 and is the narrowest of the inputs.
- **Correction:** N(0.0006, 0.0065). The mean is essentially 0, so the filter confirms the vehicle is stopped; the variance drops by more than 60% relative to the prediction.
- **Parameters:** low noise everywhere means the filter starts with high confidence and keeps it; measurements here only refine an already accurate estimate.